

EE 122/ME 141 Week 10, Lecture 2: Practice state-feedback control design

Instructor: Ayush Pandey

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1 Announcements

- Project problem statement out now. First milestone due April 12, 2026.

2 Learning Objective

1. Practice the design of state-feedback controllers for 1-D and 2-D systems.
2. Connect physical closed-loop behavior to desired poles.
3. Understand the role of the feedforward gain k_f when the goal is reference tracking instead of only regulation.

In this lecture, we will walk through three examples of state-feedback control design. The main theme in the lecture is that the physics of the system guides our choice of desired poles, which then leads us to the control laws.

3 Example 0: first-order step response

Before starting the examples, we review why the phrase “reach 63% of the final value in one time constant” appears so often in first-order systems. Consider a stable first-order closed-loop transfer function written as

$$G_{cl}(s) = \frac{b}{s + a}, \quad a > 0.$$

If the reference input is a unit step,

$$R(s) = \frac{1}{s},$$

then the output is

$$Y(s) = G_{cl}(s)R(s) = \frac{b}{s(s+a)}.$$

Using partial fractions, we can write that

$$\frac{b}{s(s+a)} = \frac{A}{s} + \frac{B}{s+a}$$

which implies

$$b = A(s+a) + Bs.$$

We get that $A = \frac{b}{a}$ and $B = -\frac{b}{a}$. So,

$$Y(s) = \frac{b}{a} \left(\frac{1}{s} - \frac{1}{s+a} \right).$$

Taking the inverse Laplace transform,

$$y(t) = \frac{b}{a} (1 - e^{-at}),$$

and the final value is

$$y(\infty) = \frac{b}{a},$$

so the response can be written as

$$y(t) = y(\infty) (1 - e^{-at}).$$

At the time

$$t = \frac{1}{a},$$

we get

$$y\left(\frac{1}{a}\right) = y(\infty) (1 - e^{-1}) \approx 0.63 y(\infty).$$

This means that for a first-order system with pole at $-a$, the time constant is $1/a$, and the output reaches about 63% of its final value after one time constant.

Pop Quiz 3.1: Check your understanding!

If we want a first-order closed-loop response to reach 63% of its final value in 1 second, what should the closed-loop pole be?

Solution on page [9](#)

4 Example 1: first-order system control design

Consider the first-order system

$$\dot{x} = -0.5x + u, \quad y = 2x.$$

We use this example to practice both state-feedback design and the derivation of the feedforward gain k_f for reference tracking.

Step 1: write the state-space matrices

From the model, we can directly read off

$$A = [-0.5], \quad B = [1], \quad C = [2], \quad D = [0].$$

Step 2: identify the open-loop pole

For this scalar system, the open-loop pole is simply the eigenvalue of A :

$$\lambda_{ol} = -0.5.$$

So the open-loop system is stable, but it is slower than what we want.

Step 3: the system requirement

We want the closed-loop output $y(t)$ to reach 63% of its final value in 1 second when a step reference is applied. From the popquiz above, we see that the desired

closed-loop pole is

$$\lambda_{cl} = -1.$$

Step 4: choose the control law

Since we want to track a reference, we use

$$u = -Kx + k_f r.$$

Substituting into the plant gives

$$\dot{x} = -0.5x + (-Kx + k_f r).$$

Therefore,

$$\dot{x} = (-0.5 - K)x + k_f r.$$

The closed-loop pole is $-0.5 - K$. To place this pole at -1 , we solve $-0.5 - K = -1$. Hence,

$$K = 0.5.$$

Step 5: derive the feedforward gain k_f

Now we derive k_f step by step rather than quoting a formula. At steady state for a constant reference r , we have $\dot{x} = 0$. So the closed-loop state equation becomes

$$0 = (-0.5 - K)x_{ss} + k_f r.$$

The output is

$$y = 2x,$$

so perfect tracking requires

$$y_{ss} = r.$$

Since $y_{ss} = 2x_{ss}$, this means

$$2x_{ss} = r \implies x_{ss} = \frac{r}{2}.$$

Substitute this into the steady-state equation:

$$0 = (-0.5 - K)\frac{r}{2} + k_f r.$$

Assuming $r \neq 0$, divide by r :

$$0 = -\frac{0.5 + K}{2} + k_f.$$

Therefore,

$$k_f = \frac{0.5 + K}{2}.$$

Using $K = 0.5$, we get

$$k_f = \frac{0.5 + 0.5}{2} = \frac{1}{2} = 0.5.$$

So the controller is

$$u = -0.5x + 0.5r.$$

Step 6: write the closed-loop system and transfer function

Substituting $K = 0.5$ and $k_f = 0.5$ into the closed-loop equation gives

$$\dot{x} = -x + 0.5r, \quad y = 2x.$$

Taking Laplace transforms with zero initial conditions,

$$sX(s) = -X(s) + 0.5R(s).$$

So

$$(s + 1)X(s) = 0.5R(s), \quad X(s) = \frac{0.5}{s + 1}R(s).$$

Since $y = 2x$,

$$Y(s) = 2X(s) = \frac{1}{s + 1}R(s).$$

Therefore, the closed-loop transfer function from r to y is

$$\frac{Y(s)}{R(s)} = \frac{1}{s + 1}.$$

This is the first-order form we wanted, with pole at -1 !! All done!

Step 7: sanity check

For a unit step reference,

$$R(s) = \frac{1}{s}.$$

Hence,

$$Y(s) = \frac{1}{s(s+1)}.$$

Using partial fractions,

$$\frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1},$$

and taking inverse Laplace transform, we get

$$y(t) = 1 - e^{-t}.$$

At $t = 1$,

$$y(1) = 1 - e^{-1} \approx 0.63.$$

So the output reaches about 63% of its final value at 1 second, as required.

5 Example 2: deriving the reachable/controllable canonical form

Consider the DC motor transfer function

$$G(s) = \frac{2}{2s+1}.$$

We first rewrite it in a form that makes the pole easier to see:

$$G(s) = \frac{1}{s+0.5}.$$

Step 1: write a state-space model in reachable canonical form

Use the approach we discussed in class: multiply numerator and denominator by an unknown function $Z(s)$:

$$G(s) = \frac{1}{s+0.5} = \frac{Z(s)}{Z(s)(s+0.5)}.$$

Then, we write $Y(s) = 1 \cdot Z(s)$ and $U(s) = Z(s)(s + 0.5)$, which gives

$$Y(s) = Z(s), \quad U(s) = Z(s)(s + 0.5).$$

Now, let's define the state as

$$x = z$$

so we get,

$$y = x, \quad u = \dot{x} + 0.5x.$$

Rearranging gives

$$\dot{x} = -0.5x + u.$$

Thus, a state-space representation in reachable canonical form is

$$A = [-0.5], \quad B = [1], \quad C = [1], \quad D = [0].$$

Step 2: identify the open-loop pole

Again, for this scalar system, the open-loop pole is

$$\lambda_{ol} = -0.5.$$

Step 3: convert the physical requirement into a desired pole

We want the closed-loop output to reach 63% of its final value in 1 second under a step input. Using the first-order derivation from the beginning of the lecture, the desired pole is

$$\lambda_{cl} = -1.$$

Step 4: compute the state-feedback gain

We get $K = 0.5$, same as before.

6 Practice problem (take home): second-order system control design

Consider the second-order system

$$\dot{x} = Ax + Bu, \quad y = Cx + Du$$

with

$$A = \begin{bmatrix} -2 & -13 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad D = 0.$$

Use `step(ss(A,B,C,D))` to plot the step response of the system. Find the open-loop poles and choose desired poles to oscillate more slowly than the open-loop system. Write down the desired characteristic polynomial, and obtain K by comparing it to the characteristic polynomial of the closed-loop system.

Pop Quiz Solutions

Pop Quiz 3.1: Solution(s)

Since the time constant must be 1, we need

$$\frac{1}{a} = 1 \quad \implies \quad a = 1.$$

Therefore, the desired pole is

$$s = -1.$$